

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Case Study – GROUP 1
(USN ending with 0, 3, 5, 7)

Course: Database Management System
 Sem / Sec: 4th (H)
 Release Date: 30/04/2026

Course code: 24CSK44
 Max. marks: 10
 Due Date: 06/05/2026

Q#	Question	RBT Level	COs	POs & PSOs	Marks
1.	A banking application processes a transaction where ₹15,000 is transferred from one account to another. During the process, the amount is deducted from the sender's account, but due to a system crash, it is not credited to the receiver's account. The database is left in an inconsistent state. Analyze this situation in terms of ACID properties. Which property is being violated, and how should the database system handle such failures to maintain correctness?	L3,L4	CO5	PO1, PO2, PO3, PO4, PS01, PS02	10
2.	In an online ticket booking system, two users simultaneously attempt to book the last available seat. Both transactions read the available seats as "1" and proceed to confirm the booking, resulting in double booking. Identify the isolation issue in this scenario. How can database isolation levels prevent such anomalies, and which level would be most appropriate here?				

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Case Study – GROUP 2
(USN ending with 1, 6, 8)

Course: Database Management System
 Sem / Sec: 4th (H)
 Release Date: 30/04/2026

Course code: 24CSK44
 Max. marks: 10
 Due Date: 06/05/2026

Q#	Question	RBT Level	COs	POs & PSOs	Marks
1.	In a payroll system, Transaction T1 updates an employee's salary but has not yet committed. Meanwhile, Transaction T2 reads this updated salary and calculates bonuses based on it. Later, T1 fails and is rolled back, but T2 has already committed its results. Examine whether this schedule is recoverable. What problems arise from such execution, and how can recoverable or cascadeless schedules avoid this issue?	L3,L4	CO5	PO1, PO2, PO3, PO4, P011, PS01, PS02	10
2.	In a hospital management system, Transaction T1 locks a patient's record to update diagnosis details. At the same time, Transaction T2 attempts to access the same record for billing purposes but must wait until T1 completes its operation and releases the lock. Explain how the Two-Phase Locking protocol operates in this scenario. Identify the growing and shrinking phases and discuss how this mechanism ensures serializability.				

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Case Study – GROUP 3
(USN ending with 2, 4, 9)

Course: Database Management System
 Sem / Sec: 4th (H)
 Release Date: 30/04/2026

Course code: 24CSK44
 Max. marks: 10
 Due Date: 06/05/2026

Q#	Question	RBT Level	COs	POs & PSOs	Marks
1.		L3,L4	CO5	PO1, PO2, PO3, PO4,P011,PS01,PS02	10
2.					